

SIGNAL SIMULATION AND CABLE COMPARISON

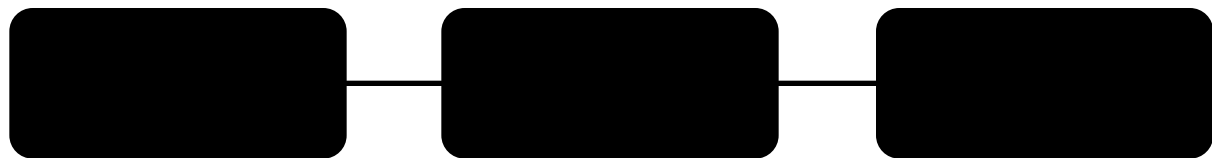
Cisco Packet Tracer Activity | Wired vs. Wireless Media

OBJECTIVE

Simulate data transmission using different physical media and compare wired and wireless network performance through connectivity testing, observations, and reflection.

TOPOLOGY A — WIRED NETWORK

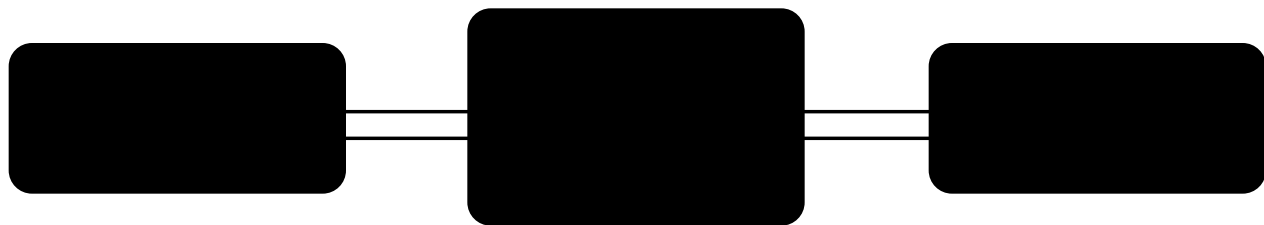
Copper Straight-Through Cable



Devices: two PCs and one Cisco 2960 switch. The PCs are connected using copper straight-through Ethernet cables.

TOPOLOGY B — WIRELESS NETWORK

Wireless 802.11 connection



Devices: two laptops and one WRT300N wireless router. Both laptops connect to the router through Wi-Fi.

IP CONFIGURATION

Topology	Device	IPv4 Address	Subnet Mask	Medium
A – Wired	PC0	192.168.10.10	255.255.255.0	Copper
A – Wired	PC1	192.168.10.20	255.255.255.0	Copper
B – Wireless	Laptop0	192.168.20.10	255.255.255.0	Wi-Fi
B – Wireless	Laptop1	192.168.20.20	255.255.255.0	Wi-Fi

CONNECTIVITY TEST

Topology A: From PC0 → Desktop → Command Prompt, run `ping 192.168.10.20`.

Topology B: From Laptop0 → Desktop → Command Prompt, run `ping 192.168.20.20`.

Expected result: Both topologies should produce successful replies when the IP settings, cables/wireless association, and interfaces are correctly configured.

OBSERVATION AND COMPARISON

Criteria	Wired – Copper	Wireless – Wi-Fi
Transmission delay	Typically stable and low	Can vary with interference and signal strength
Packet loss	Usually low in a simple LAN	May occur due to interference, distance, or weak signal
Medium used	Copper Ethernet cable	Radio waves / Wi-Fi
Mobility	Limited by cable length	High mobility within wireless coverage
Reliability	Generally more consistent	More sensitive to environmental conditions

These are expected characteristics for the media comparison. Actual delay and packet-loss values should be recorded from the student's Packet Tracer simulation if the activity requires measured values.

PING RESULTS

Topology	Source	Destination	Expected Result	Recorded Result
A – Wired	PC0	192.168.10.20	Successful	Insert Packet Tracer screenshot/result
B – Wireless	Laptop0	192.168.20.20	Successful	Insert Packet Tracer screenshot/result

SCREENSHOTS TO INCLUDE IN THE FINAL SUBMISSION

1. Topology A in Cisco Packet Tracer showing two PCs connected to the 2960 switch with copper cables.
2. Topology B showing two laptops associated with the WRT300N wireless router.
3. IP configuration of the devices.
4. Successful ping result for the wired topology.
5. Successful ping result for the wireless topology.
6. Any Packet Tracer simulation/observation showing delay or packet behavior, if available.

SHORT REFLECTION

Which physical medium performed better and why?

For this simple Packet Tracer comparison, the wired copper connection is expected to perform better in terms of consistency and reliability. Ethernet provides a direct physical connection between the PCs and the switch, so it is less affected by wireless interference, signal strength, and distance. The wireless topology provides the advantage of mobility and eliminates the need for Ethernet cables, but its performance can vary depending on signal conditions. Therefore, wired media is generally preferable when stable and predictable connectivity is the priority, while wireless is more useful when mobility and flexible device placement are important.

CONCLUSION

The activity demonstrates how the choice of transmission medium affects network connectivity. Both wired and wireless networks can successfully transmit data when properly configured. The main difference is that wired Ethernet generally offers more consistent connectivity, while wireless networking offers greater mobility and convenience.

Note: The diagrams and expected results in this document are prepared as a visual guide. Actual Packet Tracer screenshots and measured ping observations should be inserted before submission to provide evidence of the live simulation.